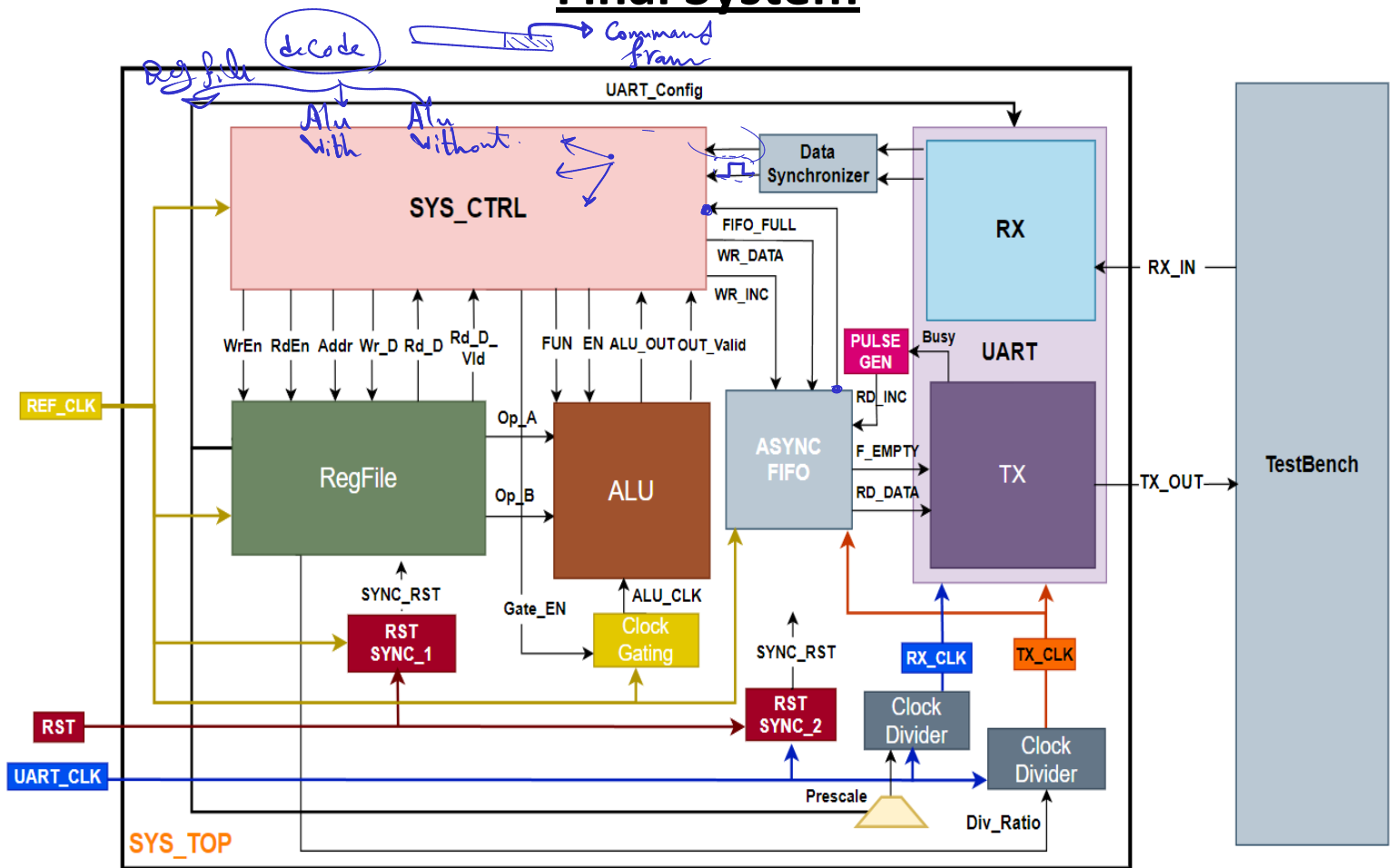


Final System



• Description: -

1 . This system contains 10 blocks: -

1) Clock Domain 1 (REF_CLK)

- RegFile
- ALU
- Clock Gating
- SYS_CTRL

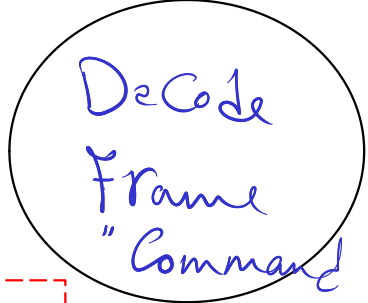
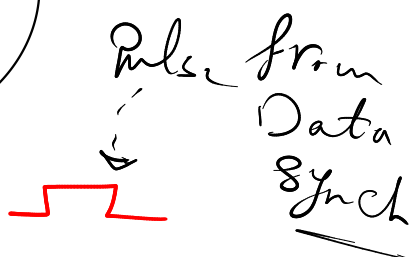
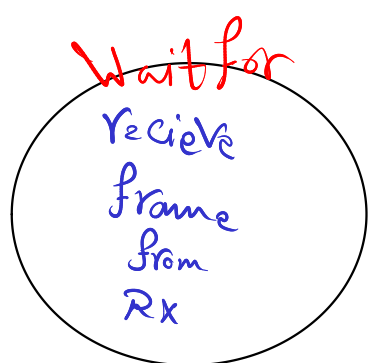
• Clock Domain 2 (UART_CLK)

- UART_TX
- UART_RX
- PULSE_GEN
- Clock Dividers

• Data Synchronizers

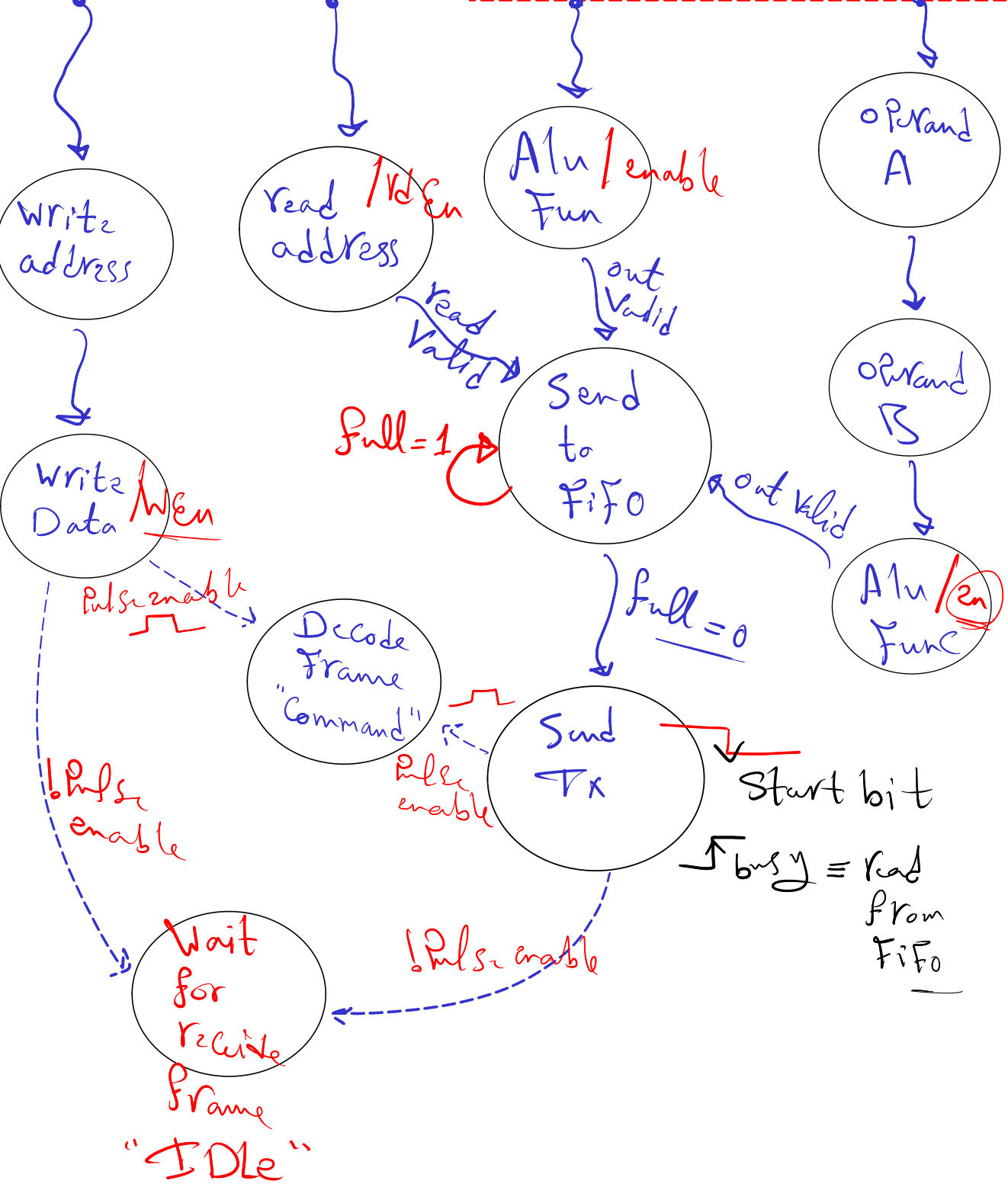
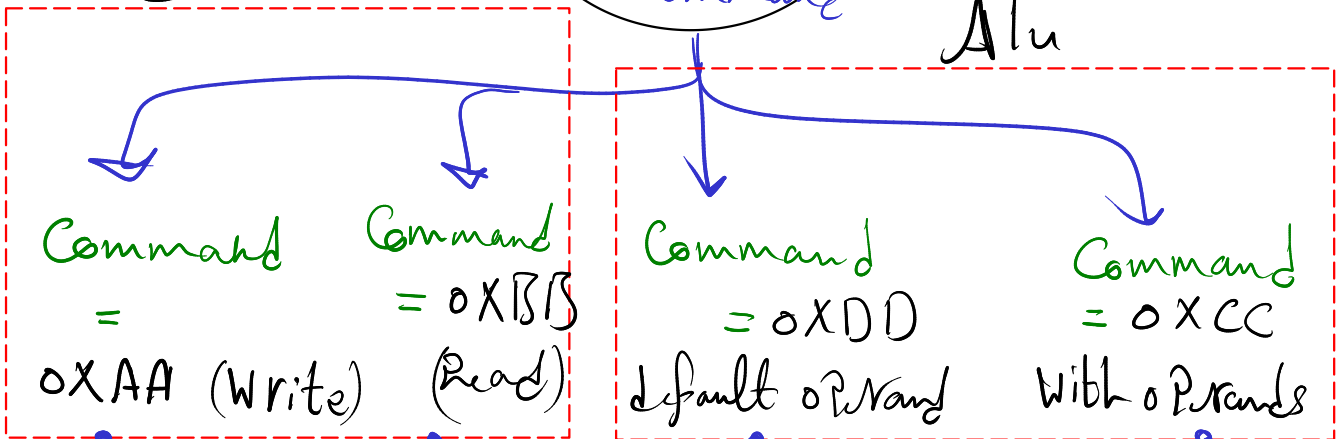
- RST Synchronizer
- Data Synchronizer
- ASYNC FIFO

1. Assume
When received
the Alu func
frame
at this state
Alu operation
is done
as En will be
active



Reg File

Alu



- **Reserved Registers Description: -**

1) REG0 (Address: 0x0)

ALU Operand A

2) REG1 (Address: 0x1)

ALU Operand B

3) REG2 (Address: 0x2)

UART Config

REG2[0]: Parity Enable
(Default = 1)

REG2[1]: Parity Type
(Default = 0)

REG2[7:2]: Prescale
(Default = 32)

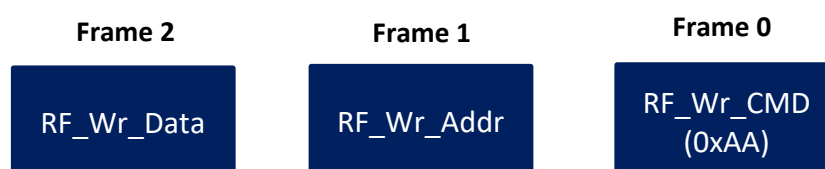
4) REG3 (Address: 0x3)

Div Ratio

REG3[7:0]: Division ratio
(Default = 32)

Introduction

- The system is responsible to do some operation based on the received **commands** from the master through **UART_RX** interface, once the operation is done, the system is responsible to send the result to the master through **UART_TX** interface.
- Supported Operations: -
 1. **ALU Operations:** -
 - Addition
 - Subtraction
 - Multiplication
 - Division
 - AND
 - OR
 - NAND
 - NOR
 - XOR
 - XNOR
 - CMP: A = B
 - CMP: A > B
 - SHIFT: A >> 1
 - SHIFT: A << 1
 2. **Register File Operations**
 - Register File Write
 - Register File read
- Supported Commands: -
 1. **Register File Write command (3 frames)**



2. Register File Read command (2 frames)



3. ALU Operation command with operand (4 frames)



4. ALU Operation command with No operand (2 frames)



System Specifications: -

- Reference clock (REF_CLK) is 50 MHz
- UART clock (UART_CLK) is 3.6864 MHz
- Clock Divider is always on (clock divider enable = 1)

Sequence of Operation (Must include in the testbench): -

- Initially configuration operations are performed through Register file write operations in addresses (0x2, 0x3).
- The Master (Testbench) start to send different commands (RegFile Operations, ALU operations)
- Our system will receive the command frames through UART_RX, it sent to the SYS_CTRL block to be processed
- Once the operation of the command is performed using ALU/RegFile, SYS_CTRL sends the result to the master through UART_TX
- Register File Address Range for normal write/read operations (From 0x4 to 0x15)
- Register File Addresses reserved for configurations and ALU operands (From 0x0 to 0x3)